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USACE / NAVFAC / AFCEC UFGS-26 29 23 (February 2026)

Preparing Activity: NAVFAC

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Superseding  
UFGS-26 29 23 (February 2020)

## UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated January 2026

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#### SECTION 26 29 23

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02/26

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### SECTION 26 29 23

#### ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS 02/26

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NOTE: This guide specification covers the requirements for ASD (also referred to as variable frequency drive (VFD)) for motors rated up to 575 volts, for use on electric power systems of 600 volts or less, 50/60 hertz.

Adhere to [UFC 1-300-02](#) Unified Facilities Guide Specifications (UFGS) Format Standard when editing this guide specification or preparing new project specification sections. Edit this guide specification for project specific requirements by adding, deleting, or revising text. For bracketed items, choose applicable item(s) or insert appropriate information.

Coordinate this guide specification with a mechanical designer for HVAC ASD application.

Military standard requirements for parent equipment with motors controlled via ASDs also apply to the ASDs. Examples include petroleum, oil and lubricant (POL) systems and Mobile/Tactical generators with ASDs hardening as a part of the generator equipment assembly.

Remove information and requirements not required in respective project, whether or not brackets are present.

Comments, suggestions and recommended changes for this guide specification are welcome and should be submitted as a [Criteria Change Request \(CCR\)](#).

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NOTE: Pulse width modulated (PWM) is the predominant type of adjustable speed drive (ASD). Other ASD types include current source inverter (CSI), voltage source inverter (VSI), and flux

vector drive (FVD). For guidance of the proper application of ASD's and installation, refer to Appendix B of UFC 3-520-01. Since the carrier-frequency pulse output voltage of a PWM cause rapid rise times in these pulses, the transmission line effects must be considered. The resulting voltage can produce overvoltages equal to twice the DC bus voltage or up to 3.1 times the rated line voltage, putting high stress on the cable and motor windings, and eventual insulation failure. Use ASD manufacturer recommended cable type and maximum safe cable distances before using the external protection devices. Use motors designed for use with ASD.

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NOTE: The use of power capacitors for power factor correction and/or the use of surge protection capacitors on the load side of an electronic control connected to an induction motor is not recommended; damage to the control may occur. Contact the control vender for suitability.

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## PART 1 GENERAL

### 1.1 REFERENCES

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NOTE: This paragraph is used to list the publications cited in the text of the guide specification. The publications are referred to in the text by basic designation only and listed in this paragraph by organization, designation, date, and title.

Use the Reference Wizard's Check Reference feature when you add a Reference Identifier (RID) outside of the Section's Reference Article to automatically place the reference in the Reference Article. Also use the Reference Wizard's Check Reference feature to update the issue dates.

References not used in the text will automatically be deleted from this section of the project specification when you choose to reconcile references in the publish print process.

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The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

EN 61800-3

(2017) Requirements for the Control of  
Electromagnetic Interference  
Characteristics of Subsystems and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 519 (2022) Standard for Harmonic Control in Electrical Power Systems
- IEEE C62.41.1 (2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
- IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

- IEC 61000-3-12 (2012) Electromagnetic Compatibility (EMC) - Part 3-12: Limits - Limits for harmonic currents produced by equipment connected to public low-voltage systems with input current >16 A and ≤75 A per phase

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
- NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements
- NEMA ICS 3.1 (2019) Guide for the Application, Handling, Storage, Installation and Maintenance of Medium-Voltage AC Contactors, Controllers and Control Centers
- NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures
- NEMA ICS 7 (2020) Adjustable-Speed Drives
- NEMA ICS 7.2 (2021) Application Guide for AC Adjustable Speed Drive Systems
- NEMA ICS 61800-2 (2005) Adjustable Speed Electrical Power Drive Systems Part 2: General Requirements - Rating Specifications for Low Voltage Adjustable Frequency A.C. Power Drive Systems
- NEMA MG 00001 (2024) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 70 (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15

Radio Frequency Devices

UL SOLUTIONS (UL)

UL 489

(2025) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

UL 61800-5-1

(2022) Adjustable Speed Electrical Power Drive Systems - Part 5-1: Safety Requirements - Electrical, Thermal and Energy

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SYSTEM DESCRIPTION

1.3.1 Performance Requirements

1.3.1.1 Electromagnetic Interference Suppression

Computing devices, as defined by 47 CFR 15 and EN 61800-3 rules and regulations, must be certified to comply with the requirements for class A computing devices and labeled.

1.3.1.2 Electromechanical and Electrical Components

Ensure electrical and electromechanical components of the Adjustable Speed Drive (ASD) do not cause electromagnetic interference to adjacent electrical or electromechanical equipment while in operation.

1.3.2 Electrical Requirements

1.3.2.1 Power Line Surge Protection

IEEE C62.41.1 and IEEE C62.41.2, IEEE 519, IEC 61000-3-12 Control panel must have surge protection, included within the panel to protect the unit from damaging transient voltage surges. Surge protective device must be mounted near the incoming power source and properly wired to all three phases and ground. Fuses must not be used for surge protection.

1.3.2.2 Sensor and Control Wiring Surge Protection

I/O functions as specified must be protected against surges induced on control and sensor wiring installed outdoors and as shown. Test the inputs and outputs in both normal mode and common mode using the following two waveforms:

- a. A 10 microsecond by 1000 microsecond waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.
- b. An 8 microsecond by 20 microsecond waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

## 1.4 SUBMITTALS

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**NOTE:** Review Submittal Description (SD) definitions in Section 01 33 00 SUBMITTAL PROCEDURES and edit the following list, and corresponding submittal items in the text, to reflect only the submittals required for the project. The Guide Specification technical editors have classified those items that require Government approval, due to their complexity or criticality, with a "G." Generally, other submittal items can be reviewed by the Contractor's Quality Control System. Only add a "G" to an item if the submittal is sufficiently important or complex in context of the project.

For Army projects, fill in the empty brackets following the "G" classification, with a code of up to three characters to indicate the approving authority. Codes for Army projects using the Resident Management System (RMS) are: "AE" for Architect-Engineer; "DO" for District Office (Engineering Division or other organization in the District Office); "AO" for Area Office; "RO" for Resident Office; and "PO" for Project Office. Codes following the "G" typically are not used for Navy and Air Force projects.

The "S" classification indicates submittals required as proof of compliance for sustainability Guiding Principles Validation or Third Party Certification and as described in Section 01 33 00 SUBMITTAL PROCEDURES.

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Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Schematic Diagrams; G, [\_\_\_\_\_]

Interconnecting Diagrams; G, [\_\_\_\_\_]

Installation Drawings; G, [\_\_\_\_\_]

As-Built Drawings; G, [\_\_\_\_\_]

### SD-03 Product Data

Adjustable Speed Drives; G, [\_\_\_\_\_]

Wires and Cables

## Equipment Schedule

### SD-06 Test Reports

#### ASD Test

#### Performance Verification Tests

#### Endurance Test

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**NOTE: Choose a bracketed SD-07 if paragraph ASD  
TEST will use an option with engaging qualified  
testing agency's field supervisor.**  
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### [ SD-07 Certificates

Testing Agency's Field Supervisor NETA Certificate; G, [\_\_\_\_]

### ] SD-08 Manufacturer's Instructions

#### Installation instructions

### SD-09 Manufacturer's Field Reports

ASD Test Plan; G, [\_\_\_\_]

#### Standard Products

### SD-10 Operation and Maintenance Data

Adjustable Speed Drives, Data Package 4

## 1.5 QUALITY ASSURANCE

### 1.5.1 Schematic Diagrams

Submit diagrams showing circuits and device elements for each replaceable module. Schematic diagrams of printed circuit boards are permitted to group functional assemblies as devices, provided that sufficient information is provided for government maintenance personnel to verify proper operation of the functional assemblies.

### 1.5.2 Interconnecting Diagrams

Show interconnections between equipment assemblies, and external interfaces, including power and signal conductors. Include for enclosures and external devices.

### 1.5.3 Installation Drawings

Show floor plan of each site, with ASD's and motors indicated. Indicate ventilation requirements, adequate clearances, and cable routes. Submit drawings for government approval prior to equipment construction or integration. Immediately record modifications to original drawings made during installation for inclusion into the **as-built drawings**.



#### 1.5.4 Equipment Schedule

Provide schedule of equipment supplied. Schedule must provide a cross reference between manufacturer data and identifiers indicated in shop drawings. Schedule must include the total quantity of each item of equipment supplied and data indicating compatibility with motors being driven. For complete assemblies, such as ASD's, provide the serial numbers of each assembly, and a sub-schedule of components within the assembly. Provide recommended spare parts listing for each assembly or component.

#### 1.5.5 Installation Instructions

Provide installation instructions issued by the manufacturer of the equipment, including notes and recommendations, prior to shipment to the site. Provide operation instructions prior to acceptance testing.

#### 1.5.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

#### 1.6 DELIVERY AND STORAGE

Store delivered equipment to protect from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

#### 1.7 WARRANTY

The complete system must be warranted by the manufacturer for a period of [one year] [[\_\_\_\_\_] years]. Repair or replace any component failing to perform its function as specified and documented at no additional cost to the Government. Items repaired or replaced must be warranted for an additional period of at least one year from the date that it becomes functional again, as specified in FAR 52.246-21 Warranty of Construction.

#### 1.8 MAINTENANCE

##### 1.8.1 Spare Parts

Manufacturers provide spare parts in accordance with recommended spare parts list.

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**NOTE: Coordinate with Contracting Officer on  
whether this paragraph can be included. Edit as**

**required if additional spare parts are required for a specific project. Do not use this paragraph for Navy projects.**

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[ Provide one [\_\_\_\_\_] spare ASD of each model provided for HVAC equipment, fully programmed and ready for back-up operation when connected.

#### ]1.8.2 Operation and Maintenance Data

Provide in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Provide service and maintenance information including preventive maintenance, assembly, and disassembly procedures. Include electrical drawings from electrical general sections. Provide additional information necessary to provide complete operation, repair, and maintenance information, detailed to the smallest replaceable unit. Include copies of as-built submittals. Provide routine preventative maintenance instructions, and equipment required. Provide instructions on how to modify program settings, and modify the control program. Provide instructions on drive adjustment, trouble-shooting, and configuration. Provide instructions on process tuning and system calibration.

#### 1.8.3 Maintenance Support

During the warranty period, provide on-site, on-call maintenance services by drive manufacturer's personnel on the following basis: The service must be on a per-call basis with 36 hour response. Contractor is responsible for the maintenance of all hardware and software of the system during the warranty period. Various personnel of different expertise must be sent on-site depending on the nature of the maintenance service required. Costs must include travel, local transportation, living expenses, and labor rates of the service personnel while responding to the service request. The provisions of this Section are not in lieu of, nor relieve the Contractor of, warranty responsibilities covered in this specification. Should the result of the service request be the uncovering of a system defect covered under the warranty provisions, all costs for the call, including the labor necessary to identify the defect, must be borne by the Contractor.

#### 1.8.4 Technical Support

Provide the ASDs with manufacturer's technical telephone support in English, readily available during normal working hours.

### PART 2 PRODUCTS

#### 2.1 ADJUSTABLE SPEED DRIVES (ASD)

Provide adjustable speed drive to control the speed of induction motor(s). The ASD must include the following minimum functions, features and ratings.

- a. Input circuit breaker in accordance with UL 489 with a minimum of 10,000 amps symmetrical interrupting capacity and door interlocked external operator.
- b. A converter stage in accordance with UL 61800-5-1 must change fixed voltage, fixed frequency, ac line power to a fixed dc voltage. The converter must utilize a full wave bridge design incorporating diode

rectifiers. Silicon Controlled Rectifiers (SCR) are not acceptable. The converter must be insensitive to three phase rotation of the ac line and must not cause displacement power factor of less than .95 lagging under any speed and load condition.

- c. An inverter stage must change fixed dc voltage to variable frequency, variable ac voltage for application to a standard NEMA MG 00001 Part 30 motor designed for use with adjustable frequency power supplies. Switch the inverter to produce a sine coded pulse width modulated (PWM) output waveform.
- d. The ASD must be capable of supplying 110 percent of rated full load current for one minute at maximum ambient temperature.

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**NOTE:** If constant torque required, modify to 150 percent of rated full load. Examples of these constant torque loads include general machinery, hoists, conveyors, printing presses, positive displacement pumps, some mixers and extruders, reciprocating compressors, as well as rotary compresses. Use inverter-duty motors (Standard MG 1 Part 31) for the constant torque loads.

Example of the variable torque loads include fans, centrifugal blowers, centrifugal pumps, propeller pumps, turbine pumps, agitators, and axial compressors. Use inverter-rated motors (Standard MG 1 Part 310) for the variable torque loads.

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- e. The ASD must be designed to operate from a [\_\_\_\_\_] volt, plus or minus 10 percent, three phase, 60 Hz supply, and control motors with a corresponding voltage rating.

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**NOTE:** Choose one of the bracketed sentences when more than an ASD inherent dynamic braking torque is required because an ASD system will produce only 10 to 15 percent of motor full-load torque as dynamic braking retarding torque.

External dynamic braking (also may be called "Shunt Regulator" and "Snubber") may be used when a shorter decelerating time is occasionally required. Dynamic braking method is releasing the motor generated energy in the form of heat through a voltage regulated switching transistor and resistor.

Regenerative braking may be used where a short deceleration time is needed, a high inertia is present, or large amounts of energy losses are undesirable. The regenerative braking method is changing the ASD DC bus energy into fixed frequency utility power.

Common DC bus tie braking may be used together with a dynamic braking as an alternative to dissipating the energy by a dynamic brake resistor when several

controls are used in a process and tied in parallel. During braking, energy is returned to the common DC bus tie and may be used by rest of controls and a resistor if not all energy is used by the controls.

Dynamic, regenerative, and common DC bus tie braking cannot provide holding torque.

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- f. Acceleration and deceleration time must be independently adjustable from one second to 60 seconds.

[Adjust decelerating time by[ providing an external dynamic braking resistor designed to meet NEMA ICS 61800-2 to be capable of decelerating six times the motor inertia with no more than 150 percent of rated current with the motor at its base speed.][ providing an ASD with a regenerative braking designed to return some of braking energy from the motor to the AC power distribution system.][ providing each of several ASD used in a process with a common DC bus tie designed to share the regenerative energy between tied in parallel controls.] ]Required deceleration time may be achieved using not only dynamic braking resistor but with other methods described in NEMA ICS 7.2-2015 paragraph 5.2.5.

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**NOTE: Modify this paragraph if constant torque required.**

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- g. Adjustable full-time current limiting must limit the current to a preset value which must not exceed 110 percent of the controller rated current. The current limiting action must maintain the V/Hz ratio constant so that variable torque can be maintained. Short time starting override must allow starting current to reach 175 percent of controller rated current to maximum starting torque.
- h. The controllers must be capable of producing an output frequency over the range of 3 Hz to 60 Hz (20 to one speed range), without low speed cogging. Over frequency protection must be included such that a failure in the controller electronic circuitry must not cause frequency to exceed 110 percent of the maximum controller output frequency selected.
- i. Minimum and maximum output frequency must be adjustable over the following ranges: 1) Minimum frequency 3 Hz to 50 percent of maximum selected frequency; 2) Maximum frequency 40 Hz to 60 Hz.
- j. The controller efficiency at any speed must not be less than 96 percent.
- k. The controllers must be capable of being restarted into a motor coasting in the forward direction without tripping.
- l. Protection of power semiconductor components must be accomplished without the use of fast acting semiconductor output fuses. Subjecting the controllers to any of the following conditions must not result in component failure or the need for fuse replacement:

- (1) Short circuit at controller output

- (2) Ground fault at controller output
- (3) Open circuit at controller output
- (4) Input undervoltage
- (5) Input overvoltage
- (6) Loss of input phase
- (7) AC line switching transients
- (8) Instantaneous overload
- (9) Sustained overload exceeding 115 percent of controller rated current
- (10) Over temperature
- (11) Phase reversal

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**NOTE: Class 10 means that the protection will trip in 10 seconds when current will be at 600 percent, Class 20 means that the protection will trip in 20 seconds when current will be at 600 percent. Class II ground-fault protection is an equipment protection in accordance with NEC. Retain isolated overload alarm contacts on overload relays if they are required for local or remote alarm indication of a tripping overload relay.**

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- m. Solid state motor overload protection must[ be included such that current exceeding an adjustable threshold must activate a 60 second timing circuit. Should current remain above the threshold continuously for the timing period, the controller will automatically shut down.][ have[ sensor in each phase,][ [Class 10] [Class 20] [Class 10/20 selectable] tripping characteristic selected to protect motor against voltage and current unbalance and single phasing,][ Class II ground-fault protection, with start and run delays to prevent nuisance trip on starting,][ analog communication module,][ [NC][NO] isolated overload alarm contact,][ external overload, reset push button].]
- n. Include slip compensation circuit that will sense changing motor load conditions and adjust output frequency to provide speed regulation of **NEMA MG 00001** Part 30 designed for use with adjustable frequency power supplies motors to within plus or minus 0.5 percent of maximum speed without the necessity of a tachometer generator.

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**NOTE: Retain the last bracketed sentences if the selected time delay may not prevent an ASD from an automatic restart after a power interruption until motor has stopped. Also retain the sentence if there will be an ASD field-selected automatic and manual bypass mode in subparagraph below starting**

with "Provide properly sized..."..

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- o. The ASD must be factory set for manual restart after the first protective circuit trip for malfunction (overcurrent, undervoltage, overvoltage or overtemperature) or an interruption of power. The ASD must be capable of being set for automatic restart after a selected time delay. If the drive faults again within a specified time period (adjustable 0-60 seconds), a manual restart will be required.[ Provide Bidirectional Autospeed Search capable of starting the ASD into rotating loads spinning in either direction and returning motor to set speed in proper direction, without causing damage to drive, motor, or load.]
- p. The ASD must include external fault reset capability. All the necessary logic to accept an external fault reset contact must be included.
- q. Provide critical speed lockout circuitry to prevent operating at frequencies with critical harmonics that cause resonant vibrations. The ASD must have a minimum of three user selectable bandwidths.

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**NOTE: Three-contactor-style bypass allows motor operation via the power converter or the bypass controller; with input isolating switch and barrier arranged to isolate the power converter input and output and permit safety testing and troubleshooting of the power converter, both energized and de-energized, while motor is operating in bypass mode.**

**If a bracketed sentence is chosen with a field-selectable automatic and manual bypass mode, retain requirements for a Bidirectional Autospeed Search in subparagraph above. with a requirement for ASD to be capable of being set for automatic restart after a selected time delay.**

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- r. Provide properly sized [NEMA][IEC] rated by-pass and isolation contactors to enable operation of motor in the event of ASD failure[ and for safety transfers motor between power converter output and bypass circuit using a field-selectable automatic and manual bypass mode]. Install mechanical and electrical interlocks between the by-pass and isolation contactors. Provide a selector switch and transfer delay timer. Motor overload and short circuit protective features must remain in use during the bypass mode.

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**NOTE: Per NEMA ICS 7.2, the ASD must limit harmonic distortion reflected onto the utility system. 5 percent impedance line reactors have the same benefits as the 3 percent impedance line reactors except that the 5 percent impedance line reactors provide maximum harmonic mitigation without adding capacitance. Investigate ASD voltage performance before using line reactors.**

Retain Harmonic Analysis Report bracketed sentence if an analysis from ASD manufacturer is required. IEC 61000-3-12, Table 4 limits the THDI (total harmonic current distortion) produced by equipment connected to public low-voltage systems with input more than 16 A and less or equal to 75 A per phase to be less than 48 percent.

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NOTE: In the last bracketed sentence, insert specific requirements in addition to or in lieu of the manufacturer provided RFI/EMI mitigating method if the method does not comply with the specified limitations.

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s. Each individual ASD must meet the following Total Harmonic Distortion (THD) requirements at the input terminals to the factory assembly of the ASD or at the load disconnecting means serving the ASD and filter assembly. These measurements should be taken with the drive set at 90 percent frequency (rpms) and the motor under a minimum of 50 percent demand.

- (1) The Voltage THD should not exceed 2.0 percent THD.
- (2) The Current THD should not exceed 15.0 percent THD.
- (3) If the standard factory ASD does not meet or exceed these requirements the factory must install appropriate equipment (Harmonic Traps, Filters, different Drive technology, etc.) to mitigate the distortion to assure performance of the VFD is within the limits.
- (4) These tests should be performed at the Manufacturers Laboratory facilities and submitted as part of the Product Data Submittals, in order to prevent the necessity of adding mitigation equipment in the field. If the requirements listed above are met, IEEE 519 will also be met.

[ t. Minimum Operating Conditions. Designed and constructed ASD's to operate within the following service conditions:

- (1) Ambient Temperature Rating: 0 to 120 degrees F.
- (2) Non-condensing relative humidity rating: less than 95 percent.
- (3) Ambient rating: Not exceed 1,006 meters 3,300 feet.

][2.1.1 ASD for Industrial Application

Provide the following operator control and monitoring devices mounted on the front panel of the ASD:

- a. Manual speed potentiometer.
- b. Hand-Off-Auto (HOA) switch.
- c. Power on light.

- d. Drive run power light.
- e. Local display[ capable of including ASD status, frequency, motor RPM, phase current, fault diagnostic in descriptive text, and all programmed parameters].

#### ][2.1.2 ASD for HVAC Application

ASDs must have the following features:

- a. A local operator control providing the following functions:
  - (1) Remote/Local operator selection with password access.
  - (2) Run/Stop and manual speed commands.
  - (3) All programming functions.
  - (4) Scrolling through all display functions.
- b. A local operator control panel with the following data displayed:
  - (1) ASD status.
  - (2) Frequency.
  - (3) Motor RPM.
  - (4) Phase current.
  - (5) Scrolling through all display functions.
  - (6) Fault diagnostics in descriptive text.
  - (7) All programmed parameters.
- c. Standard PI loop controller with input terminal for controlled variable and parameter settings.
- d. User interface terminals for remote control of ASD speed, speed feedback, and an isolated form C SPDT relay, which energizes on a drive fault condition.
- e. An isolated form C SPDT auxiliary relay which energizes on a run command.
- f. An adjustable carrier frequency with 16 KHz minimum upper limit.

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**NOTE: Line reactor is based upon the percent of  
 line impedance. Investigate ASD voltage performance  
 before using line reactors.**  
 \*\*\*\*\*

- g. A built-in or external line reactor with 3 percent minimum impedance to protect the DC bus capacitors and rectifier section diodes[, reduce power line transient voltage, line notching, DC bus over-voltage tripping and improve the inverter over-current and over-voltage



conditions].

h. Historical logging information and displays:

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**NOTE: Retain first subparagraph below if time and  
date stamping is not accomplished through the DDC  
system for HVAC.**

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- [ (1) Real-time clock with current time and date.
- ] (2) Running log of total power versus time.
- (3) Total run time.
- [ (4) Fault log, maintaining last [four][\_\_\_\_\_] faults with time and  
data stamp for each.
- ] (5) [\_\_\_\_\_] .
- [ i. The ASD must be capable of automatic control by a remote [4-20 mA][0  
to 10 VDC][\_\_\_\_\_] signal, by [BACnet][LONworks][\_\_\_\_\_] network  
command, or manually by the ASD control panel.
- ] j. ASDs must include the following operator programmable parameters:
  - (1) Upper and lower limit frequency.
  - (2) Acceleration and deceleration rate.
  - (3) Variable torque volts per Hertz curve.
  - (4) Starting voltage level.
  - (5) Starting frequency level.
  - (6) Display speed scaling.
  - (7) Enable/disable soft stall feature.
  - (8) Motor overload level.
  - (9) Motor stall level.
  - (10) Jump frequency and hysteresis band.
  - (11) PWM carrier frequency.
- ] k. ASD must have the following protective features:
  - (1) An electronic adjustable inverse time current limit with  
consideration for additional heating of the motor at frequencies  
below 45Hz, for the protection of the motor.
  - (2) An electronic adjustable soft stall feature, allowing the ASD to  
lower the frequency to a point where the motor will not exceed  
the full-load amperage when an overload ASD will automatically  
return to the requested frequency when load conditions permit.

- (3) A separate electronic stall at 110 percent ASD rated current, and a separate hardware trip at 190 percent current.
- (4) The ability to shut down if inadvertently started into a rotating load without damaging the ASD or the motor.
- (5) The ability to keep a log of a minimum of four previous fault conditions, indicating the fault type and time of occurrence in descriptive text.
- (6) The ability to sustain 110 percent rated current for 60 seconds.
- (7) The ability to shutdown safely or protect against and record the following fault conditions:
  - (a) Over current (and an indication if the over current was during acceleration, deceleration, or running).
  - (b) Over current internal to the drive.
  - (c) Motor overload at start-up.
  - (d) Over voltage from utility power.
  - (e) Motor running overload.
  - (f) Over voltage during deceleration.
  - (g) ASD over heat.
  - (h) Load and ground fault.
  - (i) Abnormal parameters or data in ASD EEPROM.

## ]]2.2 ENCLOSURES

Provide equipment enclosures conforming to NEMA 250, NEMA ICS 7, and NEMA ICS 6, with a heater if located outdoors. An HMCP device must provide the disconnecting means. The operating handle must protrude through the door, but the disconnect must not be mounted on the door. The handle must indicate ON, OFF, and tripped conditions. The handle must have provisions to accommodate a minimum of three padlocks in the OFF position. Interlocks must prevent unauthorized opening or closing of the ASD door with the disconnect handle in the ON position. The door handle interlock should have provisions to be defeated by qualified maintenance personnel.

## 2.3 WIRES AND CABLES

All wires and cables must conform to NEMA 250, NEMA ICS 7, NFPA 70.

## 2.4 NAMEPLATES

Nameplates external to NEMA enclosures must conform with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manufacturer's standard, permanent nameplates for internal areas of enclosures.

## 2.5 SOURCE QUALITY CONTROL

### 2.5.1 ASD Test Plan

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test plans.

### 2.5.2 ASD Test Report

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test reports.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Per **NEMA ICS 3.1**, install equipment in accordance with the approved manufacturer's printed installation drawings, instructions, wiring diagrams, and as indicated on project drawings and the approved shop drawings. A field representative of the drive manufacturer must supervise the installation of all equipment, and wiring.

### 3.2 GROUNDING

\*\*\*\*\*

**NOTE: Choose one of the bracketed sentences. AC power system grounding is a critical consideration. The control must be solidly grounded to the main distribution system ground. A ground common with electrical welding equipment or large current equipment (5x rating of the control) should not be used. If either of these two conditions exist, use an isolation transformer sized for the ASD control with a wye secondary neutral solidly grounded. Where more than one control is used, ground each directly to the system ground terminal, do not loop ground or install in series. Poor/improper grounding is providing nearly all electric noise issues.**

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Per **NEMA ICS 7.2**, ASD must be solidly grounded to the main distribution.

### 3.3 FIELD QUALITY CONTROL

Specified products must be tested as a system for conformance to specification requirements prior to scheduling the acceptance tests. Conduct performance verification tests in the presence of Government representative, observing and documenting complete compliance of the system to the specifications. Submit a signed copy of the test results, certifying proper system operation before scheduling tests.

#### 3.3.1 ASD Test

A proposed test plan must be submitted to the contracting officer at least 28 calendar days prior to proposed testing for approval. The tests must conform to **NEMA ICS 1**, **NEMA ICS 7**, and all manufacturer's safety

regulations. The Government reserves the right to witness all tests and review any documentation. Inform the Government at least 14 working days prior to the dates of testing. Perform the ASD test[ with the assistance of a factory-authorized service representative][ engaging a qualified testing agency's field supervisor currently certified by NETA to supervise on-site testing].

### 3.3.2 Performance Verification Tests

"Performance Verification Test" plan must provide the step by step procedure required to establish formal verification of the performance of the ASD. Compliance with the specification requirements must be verified by inspections, review of critical data, demonstrations, and tests. The Government reserves the right to witness all tests, review data, and request other such additional inspections and repeat tests as necessary to ensure that the system and provided services conform to the stated requirements. Inform the Government 14 calendar days prior to the date the test is to be conducted.

### 3.3.3 Endurance Test

Immediately upon completion of the performance verification test, the endurance test must commence. The system must be operated at varying rates for not less than 192 consecutive hours, at an average effectiveness level of 0.9998, to demonstrate proper functioning of the complete PCS. Continue the test on a day-to-day basis until performance standard is met. The contractor is not allowed in the building during the endurance test. The system must respond as designed.

## 3.4 DEMONSTRATION

### 3.4.1 Training

Coordinate training requirements with the Contracting Officer. Provide video tapes, if available, of all training provided to the Government for subsequent use in training new personnel. Provide all training aids, texts, and expendable support material for a self-sufficient presentation must be provided, the amount of which to be determined by the contracting officer.

#### 3.4.1.1 Instructions to Government Personnel

Provide the services of competent instructors with minimum two-year field experience with the operation and maintenance of similar ASDs who will give full instruction to designated personnel in operation, maintenance, calibration, configuration, and programming of the complete control system. Orient the training specifically to the system installed. Instructors must be thoroughly familiar with the subject matter they are to teach. The number of training days of instruction furnished must be as specified. A training day is defined as eight hours of instruction, including two 15-minute breaks and excluding lunch time; Monday through Friday. Provide a training manual for each student at each training phase which describes in detail the material included in each training program. Provide one additional copy for archiving. Provide equipment and materials required for classroom training. Provide a list of additional related courses, and offers, noting any courses recommended. List each training course individually by name, including duration, approximate cost per person, and location of course. Unused copies of training manuals must be turned over to the Government at the end of last training session.

#### 3.4.1.2 Operating Personnel Training Program

Provide one 2-hour training session at the site at a time and place mutually agreeable between the Contractor and the Government. Provide session to train 4 operation personnel in the functional operations of the system and the procedures that personnel will follow in system operation. This training must include:

- a. System overview
- b. General theory of operation
- c. System operation
- d. Alarm formats
- e. Failure recovery procedures
- f. Troubleshooting

#### 3.4.1.3 Engineering/Maintenance Personnel Training

Accomplish the training program as specified. Training must be conducted on site at a location designated by the Government. Provide a one-day training session to train four [\_\_\_\_\_] engineering personnel in the functional operations of the system. This training must include:

- a. System overview
- b. General theory of operation
- c. System operation
- d. System configuration
- e. Alarm formats
- f. Failure recovery procedures
- g. Troubleshooting and repair
- h. Maintenance and calibration
- i. System programming and configuration

-- End of Section --